

Accelerate Boltzmann Sampling of Multimodal Distributions by Training with Log-Ratio Regularization

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1 Introduction

The Boltzmann generator based on normalizing flows (Noé et al., 2019; Wu et al., 2020) is a central tool for sampling complicated distributions. On multimodal targets, however, normalizing flows trained with the reverse KL are prone to mode collapse: modes lost during training severely degrade sample quality. A standard remedy is to replace the reverse KL with an alternative divergence that better covers the target.

Let μ denote the target distribution on $\mathbb{R}^d$ and ν the pushforward distribution of the trained flow. The most direct fix is to replace the mode-seeking reverse KL by the mass-covering forward KL

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (1)$$

which is more stable on multimodal targets. The forward KL is the expectation of the *log-ratio* $\log(\mu/\nu)$ under the target distribution μ , and therefore requires samples from μ ; these can be approximated by Annealed Importance Sampling (AIS), which transports the pushforward ν toward μ through a sequence of reweighting and resampling steps.

A representative variant is the Flow AIS Bootstrap (FAB) (Midgley et al., 2022), which uses the α -divergence with $\alpha = 2$,

$$D_2(\mu\|\nu) = \int_{\mathbb{R}^d} \frac{\mu^2(x)}{\nu(x)} dx. \quad (2)$$

FAB remains purely energy-based and is capable of discovering new modes. Since the forward KL is the limit of the α -divergence as $\alpha \rightarrow 1$, the forward KL and FAB are two instances of the same α -divergence family at different choices of α . Compared with the forward KL, FAB requires importance sampling against the more delicate distribution μ^2/ν , which can introduce training instability.

While these objectives are theoretically capable of mitigating mode collapse, they are constructed via Jensen’s inequality at the level of the full distribution, so their convergence speed and stability remain limited by the complexity of the potential landscape. Since the ultimate goal is to match ν with μ , a natural question is whether one can directly minimize the absolute value of the log-ratio $\log(\mu/\nu)$ so that the flow converges faster. A second question is whether prior knowledge of the secondary modes of μ —for example, the multiple configurations of a molecule—can be incorporated into flow training.

In this paper we propose a family of log-ratio regularizations: the XX functional, defined by

$$\text{XX}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) f\left(\left|\log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)}\right|\right) dx dy, \quad (3)$$

where $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a differentiable increasing function. We penalize the *difference* of two log-ratios rather than the absolute value of a single log-ratio because the target density μ is known only up to an unknown normalization constant; this constant cancels exactly in the difference $\log(\mu(x)/\nu(x)) - \log(\mu(y)/\nu(y))$. The

XX operates in a fundamentally different regime from standard divergence quantities such as the forward KL and the D_2 used in FAB:

1. The XX penalizes the log-ratio pointwise.

Divergence-based losses rely on Jensen’s inequality to reshape the flow globally, whereas the XX adopts a more direct strategy: it penalizes variations of the log-ratio wherever they occur. As a result, the XX is sensitive to local features of the potential landscape rather than only to its global structure.

2. The XX tolerates inaccurate target samples.

Divergence-based losses such as the KL and D_2 require accurate importance samples to remain unbiased, whereas the XX only requires the log-ratio to vanish pointwise. Consequently, inaccuracies in the approximate μ -samples barely shift the minimizer. This also makes it possible to manually inject prior knowledge of known modes to guide mode discovery without compromising accuracy.

Overall, the XX is a safe-to-use regularization that accelerates any divergence-based flow training method while offering the flexibility to incorporate prior mode information. Moreover, the XX has the same scale as the usual KL divergence, so almost no additional tuning is required. The need for samples from μ is already met by the forward KL and FAB, so no additional data-preparation cost is incurred.

2 Single-step training loss

Let $U_0(x)$ and $U_1(x)$ be the source and target potential functions on $\mathbb{R}^d$, and let $\mu_0 \propto \exp(-U_0)$ and $\mu_1 \propto \exp(-U_1)$ be the corresponding source and target distributions. Given sufficient samples from μ_0 , we aim to construct a differentiable flow map G on $\mathbb{R}^d$ such that $G_{\#}\mu_1 \approx \mu_0$, so that target samples can be obtained from $G_{\#}^{-1}\mu_0$ via importance sampling.

We first review the construction of the training loss for G under the forward KL and FAB, then derive the XX functional for training G .

2.1 Forward KL

For the KL divergence, we have the symmetry equality $\text{KL}(G_{\#}\mu_1 \parallel \mu_0) = \text{KL}(\mu_1 \parallel G_{\#}^{-1}\mu_0)$. The pushforward density of $G_{\#}^{-1}\mu_0$ is given by

$$(G_{\#}^{-1}\mu_0)(y) = \mu_0(G(y))|\det J_G(y)|, \quad (4)$$

hence the forward KL reads

$$\begin{aligned} \text{KL}(G_{\#}\mu_1 \parallel \mu_0) &= - \int_{\mathbb{R}^d} \mu_1(y) \log(G_{\#}^{-1}\mu_0)(y) dy + \text{const} \\ &= \int_{\mathbb{R}^d} \mu_1(y) \left(U_0(G(y)) - \log |\det J_G(y)| \right) dy + \text{const}. \end{aligned}$$

The forward KL training loss in terms of G is defined by

$$\mathcal{L}_{\text{KL}}[G] = \mathbb{E}_{y \sim \mu_1} \left[U_0(G(y)) - \log |\det J_G(y)| \right]. \quad (5)$$

Here, we deliberately use the inverse map G from target to source so that $\mathcal{L}_{\text{KL}}[G]$ does not involve the inverse of G , for the best efficiency of training.

The forward KL requires explicit samples from the target distribution μ_1 , which can be approximately obtained via the Annealed Importance Sampling (AIS) approach. To be specific, we connect the pushforward distribution $G_{\#}^{-1}\mu_0$ and the target μ_1 with the following $M + 1$ intermediate distributions:

$$\pi_k = (G_{\#}^{-1}\mu_0)^{1-\frac{k}{M}}(\mu_1)^{\frac{k}{M}}, \quad k = 0, 1, \dots, M. \quad (6)$$

Equivalently, their negative log-likelihood is explicitly constructed as

$$\begin{aligned} -\log \pi_k(y) &= -\left(1 - \frac{k}{M}\right) \log(G_{\#}^{-1}\mu_0)(y) - \frac{k}{M} \log \mu_1(y) \\ &= \left(1 - \frac{k}{M}\right) \left(U_0(G(y)) - \log |\det J_G(y)|\right) + \frac{k}{M} U_1(y) + \text{const.} \end{aligned} \quad (7)$$

The AIS is then constructed as:

Algorithm 1: Annealed Importance Sampling (AIS) from $G_{\#}^{-1}\mu_0$ to μ_1

Input: source and target potentials U_0 and U_1 , flow map G

We approximate μ_1 samples by importance sampling against the trained flow: - Draw $x_1, \dots, x_N \sim \mu_0$ (source); - Push forward to $y_i = G^{-1}(x_i)$, so $y_i \sim G_{\#}^{-1}\mu_0$ (detach to stop gradient); - Compute the log-weights $\log w_i = U_0(x_i) - U_1(y_i) - \log |\det J_G(y_i)|$; - Resample $\{y_i\}$ by weights w_i to obtain approximate samples from μ_1 ; - Apply a few Langevin steps on $\{y_i\}$ to obtain approximate samples of μ_1 .

Because of the resampling, the approximation to μ_1 is biased, so a large batch size is recommended; the parameter λ then becomes a trade-off between overcoming mode collapse and sampling accuracy.

References

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